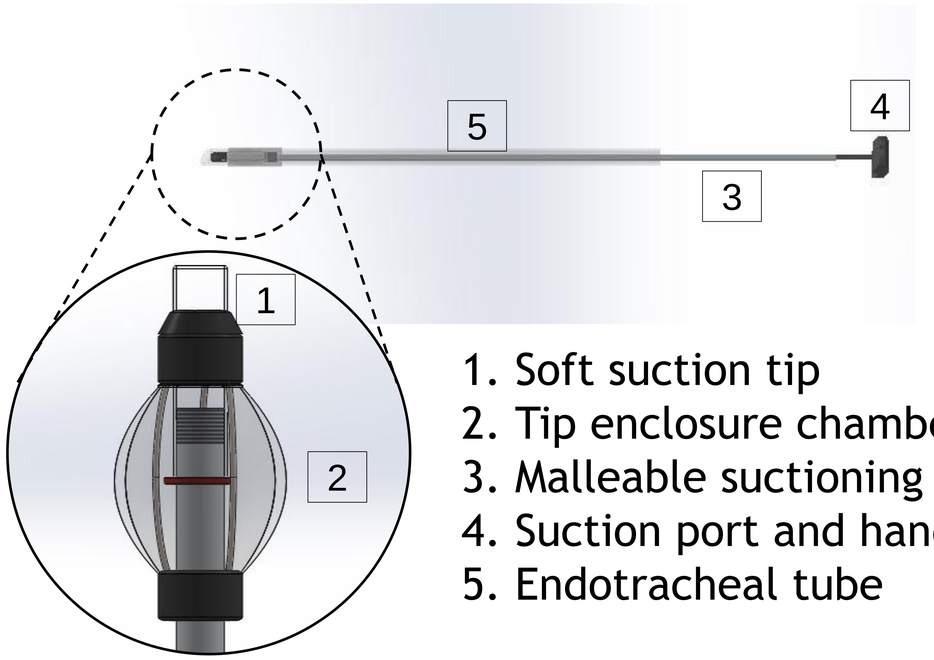
ASAP SUCTION

Problem

- Intubation: placing a breathing or endotracheal tube past the vocal folds into the main airway
- Soiled airways (i.e. blood or vomit) occurs in **29.7%** of emergency intubations
- Soiled airways decrease intubation success rates **by at least 10%**
- Health complications **increases to 63.6%** after more than three intubation attempts

Normal Airway

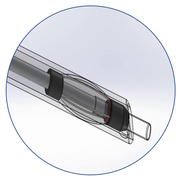
Soiled Airway



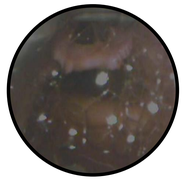
1. Soft suction tip
2. Tip enclosure chamber
3. Malleable suctioning stylet
4. Suction port and handle
5. Endotracheal tube

Steps to use ASAP Suction

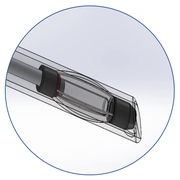
1. Insert ASAP Suction into endo-tracheal tube



2. Suction fluid, then intubate



3. Retract suction tip into the enclosure chamber



4. Extract from endo-tracheal tube

Existing Strategies

Features	Separate Suction	InScope	Simple Airway Suction Catheter
Maneuverability	✗	✗	✓
Minimal Profile	✗	✓	✓
Integrates with existing systems	✓	✗	✓
Sterile Breathing Tube	✓	✓	✗

Unique Features

- **Self-enclosing chamber** to prevent the suction tip from contaminating the internal walls of the endotracheal tube
- **Easy integration with endotracheal tube**, requiring minimal behavioural changes to intubation process

Design Evaluation

Evaluation Criteria	Weighting (%)	Intubation Concepts	
		ASAP Suction Score	Suction Integrated Laryngoscope Score
Maneuverability	25%	23%	15%
Minimal Profile	25%	23%	20%
Suction Power	18%	16%	13%
Adaptability with current systems	22%	20%	11%
Cost	10%	9%	6%
Total	100%	91%	65%
Rank		#1	#2

Projected Impact

Biomedical:

- Improved visualization of vocal folds
- Higher success rate during intubation
- Fewer complications
- Shorter stay in ICU
- Shorter stay in hospital

Economic:

- Reduced utilization of scarce hospital resources like ICU

Social:

- Higher likelihood of returning patients to their prehospital lives and families

Current Stage of Prototyping

- Completed proof of concept model
- Find material for hollow tube that is
  a) malleable
  b) stiff enough to hold its shape in the endotracheal tube, and
  c) thin-walled
- Test the device in airway mannequins